

EESC4550/5550: Practical Atmospheric Modeling (Fall 2026)
Lab 4 – Fundamental Concepts of Machine Learning
Total Score: 35

Question 1 (2+2+2=6 pts). The warming effect of CO_2 is not strictly linear. To capture the saturating behavior, where each additional ppm of CO_2 produces less additional forcing than the previous one, we estimate climate sensitivity by regressing global mean surface temperature anomaly y against atmospheric CO_2 concentration x , using the logarithmic model

$$y = a + b \ln(x).$$

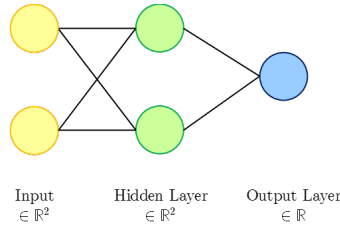
Year	CO_2 concentration (x , in ppm)	Global mean temperature anomaly (y , in $^\circ\text{C}$)
1980	340	0.20
1990	355	0.45
2000	370	0.60
2010	390	0.80
2020	415	1.10

- Fit the least-squares logarithmic regression and report a and b .
- What is the physical meaning of $b \ln(2)$?
- Evaluate the fit by computing the coefficient of determination R^2 and the root mean square error RMSE.

Question 2 (5+4=9 pts). Consider a fully-connected feedforward network with structure:

$$2 \rightarrow 2 \rightarrow 1,$$

where the input layer consists of 2 neurons, the hidden layer consists of 2 neurons, and the output layer consists of 1 neuron.



The network output is

$$\hat{y} = \sigma(\mathbf{W}^{(2)} \cdot \text{ReLU}(\mathbf{W}^{(1)} \cdot \mathbf{x} + \mathbf{b}^{(1)}) + \mathbf{b}^{(2)}),$$

where $\text{ReLU}(z) = \max(0, z)$ is applied elementwise to vectors, $\sigma(z) = (1 + e^{-z})^{-1}$, and

- $\mathbf{W}^{(1)} = \begin{pmatrix} 0 & 1 \\ 2 & -1 \end{pmatrix} \in \mathbb{R}^{2 \times 2}$, $\mathbf{b}^{(1)} = \begin{pmatrix} -1 \\ 0 \end{pmatrix} \in \mathbb{R}^2$ are the weight matrix and bias vector for the hidden layer,
- $\mathbf{W}^{(2)} = \begin{pmatrix} 1 & -1 \end{pmatrix} \in \mathbb{R}^{1 \times 2}$, $b^{(2)} = -1 \in \mathbb{R}$ are the weight matrix and bias for the output layer,
- $\mathbf{x} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \in \mathbb{R}^{2 \times 1}$ is the input vector.

- Perform a forward propagation to compute all pre-activations $\mathbf{z}^{(1)}, z^{(2)}$, post-activations $\mathbf{h}^{(1)}, h^{(2)}$, and the final prediction $\hat{y}$.
- How many trainable parameters are in this network? Are there any neurons that did not receive any weight updates? Note that they are known as the “dead neurons”.

Question 3 (1+6+4+4+2+3=20 pts). Tropospheric ozone (O_3) is a secondary pollutant formed by sunlight-driven reactions between Nitrogen Oxides (NO_x) and Volatile Organic Compounds (VOCs). However, the relationship is not simply “more $NO_2 \rightarrow$ more O_3 ”. When VOC levels are low, high NO_2 leads to ozone titration: $NO + O_3 \rightarrow NO_2 + O_2$, where NO comes from photolysis of NO_2 . Hence, if there aren't enough VOCs to recycle NO back to NO_2 without consuming O_3 , the ozone concentration would drop at very high NO_2 . Decision trees are excellent at capturing threshold effects and interactions between features. Using the table below, we predict O_3 (ppb) using two features:

- $X_1 = NO_2$ (continuous, in ppb);
- $X_2 = VOC$ (categorical, Low (L) or High (H)).

Sample	NO_2	VOC	O_3 (target)
1	10	L	30
2	20	H	45
3	30	L	40
4	40	H	55
5	50	L	50
6	60	H	70
7	70	L	45
8	80	H	75

- Compute the Sum of Squared Errors (SSE) for the target O_3 at the root.
- For the continuous feature NO_2 :
 - List all candidate split thresholds (midpoints between sorted distinct values).
 - Compute the total SSE after splitting for thresholds. (Only consider thresholds within 20-60.)
 - Identify the best threshold that gives the maximum reduction in SSE.
- For the categorical feature VOC:
 - Compute the total SSE after splitting on $\{VOC = L\}$ vs $\{VOC = H\}$.
 - Compute the reduction in SSE.
- Which feature is chosen as the root split, following the maximum SSE reduction principle? Justify your answer numerically.
- After performing the root split from (d), compute the predicted O_3 value for each child node (i.e., the mean target in each leaf).
- Given that we have a final decision tree with $depth = 2$ constructed as below, following the maximum SSE reduction principle. Here, the first split condition (intentionally left blank) follows your answer of the split threshold from (d). What is the predicted O_3 value if $NO_2 = 50$ ppb and $VOC = L$? Moreover, represent the decision tree with an equivalent piecewise function.

